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-- Engineer: Olasubomi Borishade
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-- Create Date: 05/30/2026 12:35:48 AM
-- Design Name: MAC Unit Test Bench
-- Module Name: MAC_Unit_Test_Bench - Behavioral
-- Project Name: Parameterized Matrix Multiply Engine

-- Description: Test bench for a fundamental digital circuit that
computes the
--              product of two numbers and adds the result to an
accumulator.
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
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library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
use STD.ENV.FINISH;


-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity MAC_Unit_Test_Bench is
--  Port ( );
end MAC_Unit_Test_Bench;
```

```
architecture Behavioral of MAC_Unit_Test_Bench is
Constant Input_Widths        : INTEGER   := 8;
Constant Accumulator_Widths : INTEGER   := 32;
Constant Clock_Period        : TIME      := 10 ns;

Signal Input_As              : SIGNED (Input_Widths - 1 downto 0)  :=
(others => '0');
Signal Input_Bs              : SIGNED (Input_Widths - 1 downto 0)  :=
(others => '0');

Signal Clocks                : STD_lOGIC := '0';
Signal Resets                : STD_lOGIC := '0';
Signal Enables               : STD_lOGIC := '0';
Signal Clear_Accumulators    : STD_lOGIC := '0';
Signal Outputs               : SIGNED (Accumulator_Widths - 1 downto
0) := (others => '0');

begin

MAC_Unit_Inst: entity work.Mac_Unit(Behavioral)
    Generic map (   Input_Width             => Input_Widths,
                    Accumulator_Width       => Accumulator_Widths
                    )

    Port map    (   Input_A                 => Input_As,
                    Input_B                 => Input_Bs,
                    Clock                   => Clocks,
                    Reset                   => Resets,
                    Clear_Accumulator       => Clear_Accumulators,
                    Enable                  => Enables,

                    Output                  => Outputs
                    );

    Clock_process : process
    begin
        Clocks <= '0';
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        wait for Clock_Period / 2;
        Clocks <= '1';
        wait for Clock_Period / 2;
    end process;

    Stimulus: Process
    begin

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--                                                       TEST 1
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        Resets       <= '1';
        wait for Clock_Period;
        Resets       <= '0';
        wait for Clock_Period;

        Enables      <= '1';
        Input_As     <= to_signed(3, Input_Widths);
        Input_Bs     <= to_signed(4, Input_Widths);
        wait for Clock_Period;

        Input_As     <= to_signed(0, Input_Widths);
        Input_Bs     <= to_signed(0, Input_Widths);
        Enables      <= '0';
        wait for Clock_Period;

        assert (signed(Outputs) = to_signed(12,
Accumulator_Widths))
            report "TEST 1 FAILED: Expected 12, got " &
integer'image(to_integer(signed(Outputs)))
        severity error;
```

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--                                                        TEST 2
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        Enables     <= '1';
        Input_As    <= to_signed(2, Input_Widths);
        Input_Bs    <= to_signed(3, Input_Widths);
        wait for Clock_Period;

        Clear_Accumulators <= '1';
        wait for Clock_period;

        Clear_Accumulators <= '0';
        Input_As    <= to_signed(4, Input_Widths);
        Input_Bs    <= to_signed(5, Input_Widths);
        wait for Clock_Period;

        Input_As    <= to_signed(6, Input_Widths);
        Input_Bs    <= to_signed(7, Input_Widths);
        wait for Clock_Period;

        Input_As    <= to_signed(8, Input_Widths);
        Input_Bs    <= to_signed(9, Input_Widths);
        wait for Clock_Period;

        Input_As    <= to_signed(0, Input_Widths);
        Input_Bs    <= to_signed(0, Input_Widths);
        Enables     <= '0';
```

```
        wait for Clock_Period;

        assert (signed(Outputs) = to_signed(134,
Accumulator_Widths))
            report "TEST 2 FAILED: Expected 134, got " &
integer'image(to_integer(signed(Outputs)))
        severity error;




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--                                                       TEST 3
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        Clear_Accumulators <= '1';
        wait for Clock_Period;

        assert (signed(Outputs) = to_signed(0, Accumulator_Widths))
            report "TEST 3 FAILED: Expected 0 after clear, got " &
integer'image(to_integer(signed(Outputs)))
            severity error;




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--                                                       TEST 4
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        Clear_Accumulators <= '0';
        Enables     <= '1';
        Input_As    <= to_signed(-3, Input_Widths);
        Input_Bs    <= to_signed(4, Input_Widths);
        wait for Clock_Period;

        Input_As    <= to_signed(0, Input_Widths);
        Input_Bs    <= to_signed(0, Input_Widths);
        Enables     <= '0';
        wait for Clock_Period;

        assert (signed(Outputs) = to_signed(-12,
Accumulator_Widths))
            report "TEST 4 FAILED: Expected -12, got " &
integer'image(to_integer(signed(Outputs)))
        severity error;




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--                                                           DONE
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        report "All MAC unit tests completed."
        severity note;

        finish;
    end process;
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end Behavioral;
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